

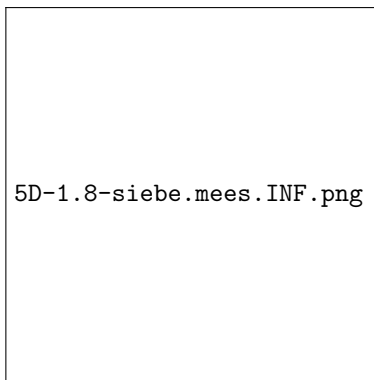
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 Studierichting: Informatica
 Oefeningenreeks 5D nr. 1.8

$$f(x) = \cos x - \sin x \text{ op } \left[0, \frac{\pi}{4}\right]$$

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} (\cos x - \sin x) dx \\ &= [\sin x + \cos x]_0^{\frac{\pi}{4}} \\ &= \left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4}\right) - (\sin 0 + \cos 0) \\ &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right) - (0 + 1) \\ &= \sqrt{2} - 1 \end{aligned}$$

$\Rightarrow$ Tekening

x	0	$\frac{\pi}{4}$
$f(x)$	1	0
$f'(x)$	—	—
$f''(x)$	—	0
$f(x)$	$\searrow$ (	$\searrow$ (



References